Midterm Test
F18CE – Multivariable Calculus
Test 3 2025/26

1 April 2026

Duration: 30 minutes

No calculators allowed

1. Find, in terms of a and b, the largest possible area of a rectangle with all its sides parallel to the coordinate axes that can be inscribed in an ellipse $(\frac{x}{a})^2 + (\frac{y}{b})^2 = 1$, $a, b > 0$. Use the method of Lagrange multiplier. **[5]**

2. Suppose D is the region enclosed between circles $x^2 + y^2 = 4$ and $x^2 + y^2 = -2y$.

(a) Sketch the region D. **[2]**

(b) Express the area of D in terms of double integrals in the cartesian coordinates. **[4]**

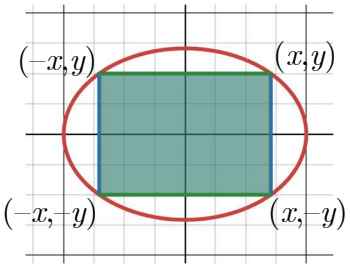
(c) Compute $\iint_D \sqrt{x^2 + y^2}\, dA$ using double integrals in the polar coordinates. **[6]**

ANSWER SCHEME

Question 1.

Let $a,b,x,y>0$, where a,b are constant. Let $f: D_f \to \mathbb{R}$ be a function defined as $f(x,y)=4xy$, where f is subject to $g(x,y)=(\frac{x}{a})^2+(\frac{y}{b})^2-1=0$, (1 mark) i.e.,

$$D_f=\left\{(x,y)\in\mathbb{R}^2:(\tfrac{x}{a})^2+(\tfrac{y}{b})^2=1, x,y>0\right\}.$$



Applying the Lagrange multiplier $\nabla(f+\lambda g)=0$: As $a,b,x,y>0$,

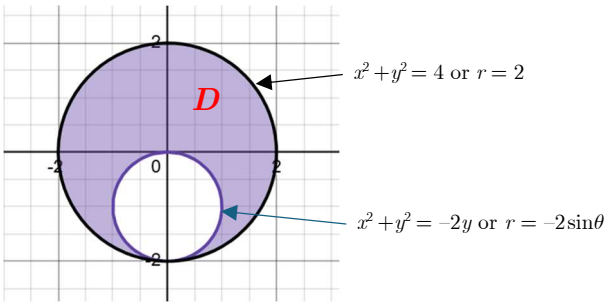
$$\begin{cases}4y+\frac{2}{a^2}\lambda x=0\\ 4x+\frac{2}{b^2}\lambda y=0\end{cases}\Rightarrow\lambda=-\frac{2a^2y}{x}=-\frac{2b^2x}{y}\Rightarrow(ay)^2=(bx)^2\Rightarrow ay=bx \quad \text{(1 mark)}$$

Substitute $ay=bx$ in $(\frac{x}{a})^2+(\frac{bx}{ab})^2=1$ to get $x=\frac{a}{\sqrt{2}}$ and $y=\frac{b}{\sqrt{2}}$. (1 mark)

Pick $x=\frac{a}{2}\sqrt{3}$, $y=\frac{b}{2}$. Then $f(\frac{a\sqrt{3}}{2},\frac{b}{2})=ab\sqrt{3}<2ab=f(\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}})$ is the max. (1 mark)

The largest possible area of a rectangle is $2ab$. (1 mark)

Question 2 (a) (2 marks)



(b) Note that the answer is EXACTLY the same if we use $r = 2\sin\theta$.

$$2\int_0^2 \int_{\sqrt{2y-y^2}}^{\sqrt{4-y^2}} 1\, dxdy + 2\int_0^2 \int_0^{\sqrt{4-y^2}} 1\, dxdy \text{ (first quadrant) (the answer is } 3\pi\text{) (2 + 2 marks)}$$

Alternative: $4\int_0^2 \int_0^{\sqrt{4-y^2}} 1\, dxdy - 2\int_0^2 \int_0^{\sqrt{2y-y^2}} 1\, dxdy$ (first quadrant)

Alternative: $2\int_{-2}^0 \int_{\sqrt{-2y-y^2}}^{\sqrt{4-y^2}} 1\, dxdy + 2\int_0^2 \int_0^{\sqrt{4-y^2}} 1\, dxdy$ (fourth quadrant)

Alternative: $4\int_0^2 \int_0^{\sqrt{4-y^2}} 1\, dxdy - 2\int_{-2}^0 \int_0^{\sqrt{-2y-y^2}} 1\, dxdy$ (fourth quadrant)

Alternative: $2\int_{-2}^0 \int_{-\sqrt{4-y^2}}^{-\sqrt{-2y-y^2}} 1\, dxdy + 2\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1\, dxdy$ (third quadrant)

Alternative: $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1\, dxdy - \int_{-2}^0 \int_{-\sqrt{-2y-y^2}}^{\sqrt{-2y-y^2}} 1\, dxdy$ (third quadrant)

Alternative: $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1\, dxdy - 2\int_{-2}^0 \int_0^{\sqrt{-2y-y^2}} 1\, dxdy$ (third quadrant)

Or by using other alternatives.

(c) $\boxed{\text{Answer } = \frac{16}{3}(\pi - \frac{2}{3}) = \frac{16}{3}\pi - \frac{32}{9} = \frac{16}{9}(3\pi - 2)}$

Since $\sqrt{x^2+y^2}\,dxdy = |r||r|drd\theta = r^2 drd\theta$, it follows that

$$2\int_0^{\pi/2}\int_{r=2\sin\theta}^{r=2} r^2 drd\theta + 2\int_0^{\pi/2}\int_0^2 r^2 drd\theta \qquad \text{(2 + 2 marks)}$$

$$= \frac{2}{3}\int_0^{\pi/2}\left(\left[r^3\right]_{2\sin\theta}^2 + \left[r^3\right]_0^2\right) d\theta \qquad \text{(1 mark)}$$

$$= \frac{16}{3}\int_0^{\pi/2}(2 - \sin^3\theta)\, d\theta$$

$$= \frac{16}{3}\left[2\theta + \cos\theta - \frac{1}{3}\cos^3\theta\right]_0^{\pi/2}$$

$$= \frac{16}{3}(\pi - \frac{2}{3}) \qquad \text{(1 mark)}$$

Alternative: $4\int_0^{\pi/2}\int_0^2 r^2 drd\theta - 2\int_0^{\pi/2}\int_0^{2\sin\theta} r^2 drd\theta = \frac{2}{3}\int_0^{\pi/2}\left(2\left[r^3\right]_0^2 - \left[r^3\right]_0^{2\sin\theta}\right) d\theta$

$$= \frac{16}{3}\int_0^{\pi/2}(2 - \sin^3\theta)\, d\theta$$

$$= \frac{16}{3}\left[2\theta + \cos\theta - \frac{1}{3}\cos^3\theta\right]_0^{\pi/2}$$

$$= \frac{16}{3}(\pi - \frac{2}{3})$$

Alternative: $4\int_{-\pi/2}^0\int_0^2 r^2 drd\theta - 2\int_{-\pi/2}^0\int_0^{-2\sin\theta} r^2 drd\theta = \frac{2}{3}\int_{-\pi/2}^0\left(2\left[r^3\right]_0^2 - \left[r^3\right]_0^{-2\sin\theta}\right) d\theta$

$$= \frac{16}{3}\int_{-\pi/2}^0(2 + \sin^3\theta)\, d\theta$$

$$= \frac{16}{3}\left[2\theta - \cos\theta + \frac{1}{3}\cos^3\theta\right]_{-\pi/2}^0$$

$$= \frac{16}{3}(\pi - \frac{2}{3})$$

Alternative: $2\int_{-\pi/2}^0\int_{r=-2\sin\theta}^{r=2} r^2 drd\theta + 2\int_{-\pi/2}^0\int_0^2 r^2 drd\theta = \frac{2}{3}\int_{-\pi/2}^0\left(\left[r^3\right]_{-2\sin\theta}^2 + \left[r^3\right]_0^2\right) d\theta$

$$= \frac{16}{3}\int_{-\pi/2}^0(2 + \sin^3\theta)\, d\theta$$

$$= \frac{16}{3}\left[2\theta - \cos\theta + \frac{1}{3}\cos^3\theta\right]_{-\pi/2}^0$$

$$= \frac{16}{3}(\pi - \frac{2}{3})$$

Or by using other alternatives.